

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 21

(1) Research Program/Project/Study Title: Production Performance of Okra (<i>Abelmoschus esculentus</i> L.) Applied with Formulated Soil Ameliorants	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao
Province:	Capiz
Region:	VI

(5) Classification Research: ☐ Basic ☒ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity Okra	(8) Discipline/Field of Specialization Agriculture
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	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 2 of 21

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted to determine the production performance of okra (*Abelmoschus esculentus* L.) applied with formulated soil ameliorants. The experiment was arranged in a Randomized Complete Block Design (RCBD) consisted of two factors with three replications. The formulated soil ameliorants (factor A) which served as treatments in the study were: A – control, B – bokashi with sheep manure, C – bokashi with hog manure and D – bokashi with chicken dung and factor B composed of cropping trials, namely: 1st cropping (1) and 2nd cropping (2). The data were subjected to the Analysis of Variance using F-test for RCBD. Differences among treatment means were compared using Least Significant Difference test (LSD), while differences between treatments were assessed using the Duncan Multiple Range Test (DMRT) and interpreted at 5% level of significance. All statistical analyses used the Statistical Tools for Agricultural Research (STAR) Nebula program developed by the International Rice Research Institute (IRRI).

The growth and yield of okra varied significantly with the application of CAFSA with different farm manures except except for plant height, biomass, weight of roots, diameter of fruit, length of fruit, and weight of marketable fruit; while cropping seasons significantly affected the growth performance of okra except the plant height, weight of roots and length of fruit.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Formulated soil ameliorant, bokashi, chicken dung, sheep manure, hog manure, cropping

- 1. Introduction** (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Okra (*Abelmoschus esculentus* L.) is a widely cultivated vegetable known for its nutritional value and versatility in culinary applications. Its production, however, can be influenced significantly by soil health and fertility. Soil ameliorants, including various organic amendments, play a crucial role in enhancing soil properties, thus boosting crop performance. This study focuses on the production performance of okra when applied with a formulated soil ameliorant containing animal manure, garden soil, compost, vermicast, rice bran, carbonized rice hull, effective microorganism inoculant, fermented plant juice, fish amino acid, indigenous microorganisms, lactic acid, molasses, and water.

Soil ameliorants are materials added to soil to improve its physical properties, enhance fertility, and increase productivity. The use of organic amendments has been widely recognized for its ability to improve soil structure, water retention, and nutrient availability (Lal, 2015). Animal manure, a common organic amendment, is rich in essential nutrients such as nitrogen, phosphorus, and potassium, which are vital for plant growth (Bhattacharyya et al., 2015). Garden soil and compost contribute to the organic matter content, which improves soil aeration and microbial activity (Eghball & Power, 1999).

Vermicast, the product of earthworm digestion, is another valuable component that enhances nutrient availability and promotes plant growth through the presence of growth-promoting substances (Arancon et al., 2004). Rice bran and carbonized rice hull are effective in improving soil texture and water holding capacity (Sombatpanit et al., 2013). Effective microorganisms (EM) inoculant, a mixture of beneficial microorganisms, is known to enhance soil microbial diversity and activity, leading to improved nutrient cycling and plant health (Higa & Parr, 1994).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 3 of 21

Fermented plant juice (FPJ) and fish amino acid (FAA) are rich in enzymes, hormones, and amino acids that can stimulate plant growth and enhance resistance to pests and diseases (Kong et al., 2008). Indigenous microorganisms (IMO) and lactic acid bacteria are beneficial in decomposing organic matter and suppressing soil-borne pathogens (Higa, 2000). Molasses, as a carbon source, provides energy for microbial activity, promoting the decomposition of organic matter and nutrient release (Subedi & Ma, 2009).

Studies have shown that integrating these components can significantly enhance soil fertility and crop yield. For instance, Arancon et al. (2004) demonstrated that the application of vermicast increased the growth and yield of tomatoes and peppers. Similarly, Higa and Parr (1994) reported that effective microorganisms improved soil quality and crop productivity. The synergistic effects of these soil ameliorants can lead to improved okra production by enhancing nutrient availability, soil structure, and microbial activity.

This study aims to evaluate the production performance of okra when applied with a formulated soil ameliorant. By integrating various organic amendments, this research seeks to provide insights into sustainable agricultural practices that can enhance okra yield and quality.

Objectives of the Study

The study aimed to find out the production performance of okra (*Abelmoschus esculentus* L.) when applied with formulated soil ameliorants. Specifically, the objectives were to:

1. Assess the growth performance of okra when applied with CAFSA containing different farm manures;
2. Determine the yield performance of okra when applied with CAFSA;
3. Evaluate the performance of okra planted in different cropping seasons;
4. Know the best CAFSA formulation in growing okra; and
5. Determine the interaction effect between CAFSA formulations and cropping seasons.

(10) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Experimental Method

The study was conducted to determine the production performance of okra (*Abelmoschus esculentus* L.) applied with formulated soil ameliorants. The experiment was arranged in a Randomized Complete Block Design (RCBD) and consisted of two factors with three replications. The CAPSU formulated soil ameliorants (CAFSA) (factor A) which served as treatments in the study were: A – Vermicast (control), B – CAFSA with sheep manure, C – CAFSA with hog manure, and D – CAFSA with chicken dung. Factor B comprised cropping season, namely: 1st cropping (February) and 2nd cropping (June). The data on the production performance of okra (*Abelmoschus esculentus* L.) applied with formulated soil ameliorants were subjected to the Analysis of Variance using F-test for RCBD. Differences among treatment means were compared using the Least Significant Difference test (LSD), while differences between treatments were assessed using the Duncan Multiple Range Test (DMRT) and interpreted at the 5% level of significance. All statistical analyses used the Statistical Tools for Agricultural Research (STAR) Nebula program developed by the International Rice Research Institute (IRRI).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 4 of 21

(11) **Results and Discussions** (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Plant Height (cm)

The data presented in Figure 1 show an interaction effect between CAFSA formulations and cropping season of okra. The tallest plant height was recorded from plants grown in CAFSA with sheep manure during the first cropping, which was comparable to CAFSA with chicken dung in the first cropping. This was followed by plants grown in CAFSA with hog manure during the first cropping, followed by those grown in vermicast during the first cropping, and found comparable with those plants grown in CAFSA with sheep manure during the second cropping. Plants grown in CAFSA with sheep manure during the second cropping produced similar height to those grown in CAFSA with hog manure and CAFSA with chicken dung, both during the second cropping. The shortest plant was recorded from those grown in vermicast, with comparable height to those grown in CAFSA with hog manure and chicken dung, all during the second cropping.

Stem Girth (mm)

The ANOVA on the stem girth (mm) of okra revealed highly significant results on the effect of both CAFSA formulations and cropping season as shown in Figures 2 and 3, respectively. The data revealed that CAFSA with sheep manure (9.62 mm) obtained the significantly highest result, which was comparable to CAFSA with chicken dung (9.19 mm). This was followed by those grown in CAFSA with hog manure (8.52 mm), with comparable stem girth to plants grown in CAFSA with chicken dung, while the control (6.74 mm) resulted in the lowest treatment mean. For the effect of cropping season, plants grown during the second cropping (9.35 mm) had significantly bigger stem girth than those grown during the first cropping (7.69 mm). There was no interaction effect between CAFSA formulations and cropping season in influencing the stem girth of okra.

Number of Leaves

The ANOVA on the number of leaves in okra showed a highly significant effect of both CAFSA formulations and cropping seasons as presented in Figures 5 and 6, respectively. CAFSA with sheep manure (8.80) produced the most number of leaves, which was comparable to CAFSA with chicken dung (8.53) and CAFSA with hog manure (8.18). Meanwhile, vermicast (7.23) produced the least number of leaves. For cropping season, the first cropping (10.14) resulted in higher leaf production compared to the second cropping (6.23). There was no interaction effect between CAFSA formulations and cropping seasons.

Width of Leaves (cm)

Figure 8 shows that the width of leaves (cm) in okra was highly significantly affected by CAFSA formulations and cropping seasons. Application of CAFSA with sheep manure (21.18 cm) resulted in the widest leaf, which was comparable to CAFSA with chicken dung (20.39 cm) and CAFSA with hog manure (19.40 cm). Meanwhile, plants applied with vermicast (15.95 cm) produced the narrowest leaf. Regarding cropping season, plants grown during the first cropping (22.64 cm) produced wider leaves compared to plants grown during

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 5 of 21

the second cropping (15.81 cm). There was no interaction effect between CAFSA formulations and cropping season.

Length of Leaf (cm)

A highly significant effect of CAFSA formulations and cropping on the length of leaf was obtained, as reflected by the ANOVA result in Figure 9. Okra applied with CAFSA with sheep manure (17.10 cm) produced the longest leaf, which was comparable to those applied with CAFSA with chicken dung (16.19 cm). Furthermore, CAFSA with hog manure (15.01 cm) was found to be comparable to CAFSA with chicken dung, while plants applied with vermicast (12.66 cm) produced the shortest leaf. On the other hand, okra planted in the first cropping (17.43 cm) produced longer leaves than those planted during the second cropping (13.45 cm). There was no interaction effect between CAFSA formulations and cropping season.

Biomass (g)

Figure 10 revealed that the biomass of okra was significantly affected by cropping season but not by CAFSA formulations. Biomass ranged from 359.53 g to 595.38 g. For the cropping season, heavier biomass was obtained from okra planted during the first cropping (287.55 g) compared to those planted during the second cropping (195.04 g). There was no interaction effect between CAFSA formulations and cropping seasons.

Length of Roots (cm)

The length of roots (cm) of okra was significantly affected by CAFSA formulations and cropping seasons, as shown in Figures 11 and 12, respectively. Okra grown in CAFSA with sheep manure (41.80 cm) had the longest root, which was comparable to CAFSA with chicken dung (33.30 cm) and CAFSA with hog manure (31.60 cm). The shortest root was recorded from okra applied with vermicast (25.81 cm), which was comparable to CAFSA with chicken dung and hog manure. Okra grown during the second cropping (38.15 cm) had longer roots compared to okra grown during the first cropping (28.11 cm). There was no interaction effect between CAFSA formulations and cropping seasons.

Weight of Roots (g)

The weight of roots of okra was not affected by the application of CAFSA formulations and cropping seasons. For CAFSA formulations, means ranged from 28.57 g to 39.66 g. For cropping seasons, means ranged from 33.59 g to 33.93 g. Likewise, no interaction effect between CAFSA formulations and cropping seasons was observed.

Diameter of Fruit (mm)

The ANOVA for the diameter of fruit showed no significant result for the effect of CAFSA formulations and a significant effect of cropping seasons, as shown in Figure 13. Means ranged from 16.01 mm to 16.29 mm for CAFSA formulations. Okra planted during the second cropping (16.77 mm) produced bigger fruits compared to those planted during the first cropping (15.48 mm). There was no interaction effect between CAFSA formulations and cropping seasons.

Length of Fruit (cm)

The ANOVA on the length of roots (cm) in okra revealed no significant effects on the

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 6 of 21

cropping trials of the experiment or when applied with formulated soil ameliorants. CAFSA with sheep manure (13.31 cm) obtained the shortest length of roots, while the control, CAFSA with hog manure, and CAFSA with chicken dung (13.39 cm) attained the longest length of roots. Meanwhile, cropping trials ranged from 13.07 cm in the first cropping to 13.66 cm in the second cropping. Additionally, there was no interaction effect between formulated soil ameliorants and cropping trials in the study.

Weight of Marketable Fruit (g)

The data in Figure 14 indicate that the weight of marketable fruit (g) of okra was not affected by the different CAFSA formulations but was significantly affected by cropping seasons. Means ranged from 16.62 g to 20.04 g for CAFSA formulations. Regarding cropping seasons, okra grown during the second cropping (19.33 g) had heavier fruit compared to those grown during the first cropping (16.68 g). An absence of interaction effect between CAFSA formulations and cropping seasons was observed.

Yield (T/ha)

The effect of CAFSA formulations and cropping season on okra yield (T/ha) was highly significant, as indicated in Figures 14 and 15, respectively. Okra applied with CAFSA formulation containing sheep manure (0.06 T/ha) and chicken dung (0.06 T/ha) had the highest yield. Furthermore, CAFSA formulation with hog manure (0.05 T/ha) was found to be comparable to CAFSA with chicken dung. Okra applied with vermicast (0.04 T/ha) had the lowest yield, which was comparable to the yield of those applied with CAFSA formulations containing hog manure. Okra grown during the second cropping produced higher yield (0.06 T/ha) compared to those grown during the first cropping (0.04 T/ha). There was no interaction effect noted between CAFSA formulations and cropping seasons.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

The findings of the study resulted in the following conclusions being drawn:

1. The growth performance of okra varied significantly with the application of CAFSA with different formulations, except for plant height, biomass, weight of roots, diameter of fruit, length of fruit, and weight of marketable fruit.
2. Cropping seasons significantly affected the growth performance of okra, except for plant height, weight of roots, and length of fruit.
3. In terms of yield performance, okra applied with CAFSA containing sheep manure resulted in the highest harvest among treatments.
4. Planting okra during the second cropping season, in the month of June, produced better yield.
5. An interaction effect between the CAFSA formulations and cropping seasons existed, influencing plant height only.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 7 of 21

Recommendations

The results of the research led to the following recommendations:

1. Use CAFSA with sheep manure to achieve a high yield for okra.
2. Use bokashi to promote better growth of okra.
3. Conduct similar studies on other high-value crops.
4. Subject the different CAFSA formulations to nutrient analysis.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

References

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	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 8 of 21

APPENDICES

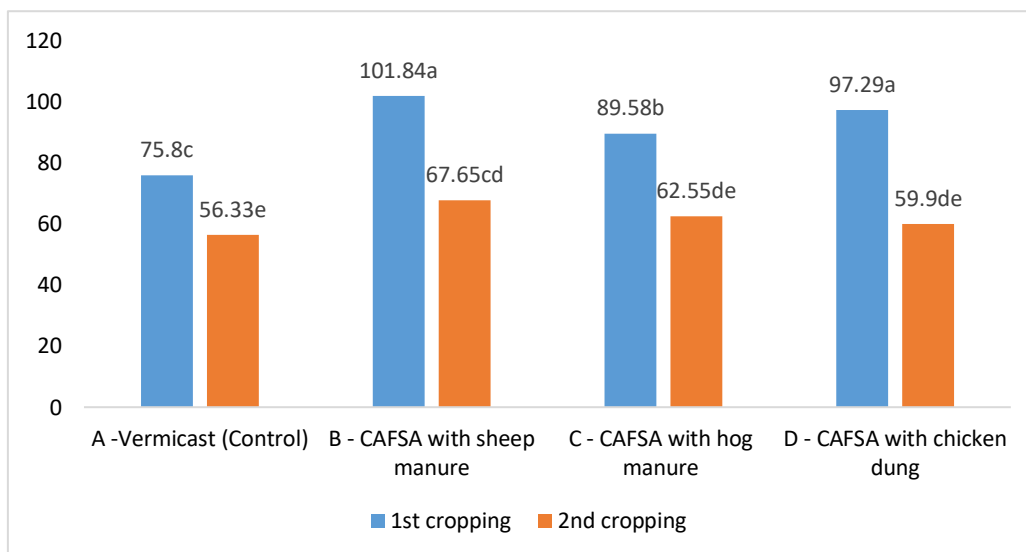


Figure 1. Interaction effect of formulated soil ameliorants and cropping seasons on the plant height (cm) of okra.

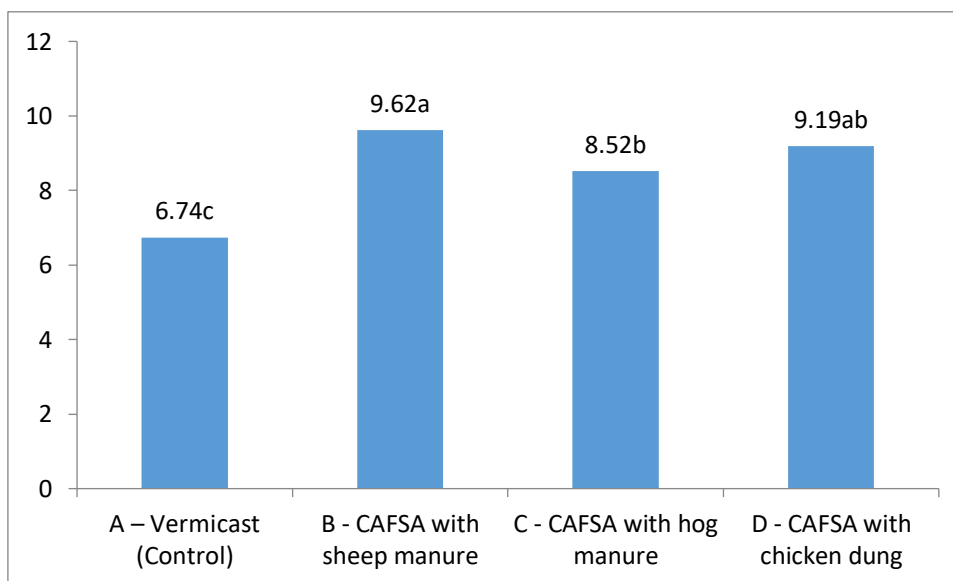


Figure 2. Stem girth (mm) of okra across two cropping season as affected by formulated soil ameliorants.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 9 of 21

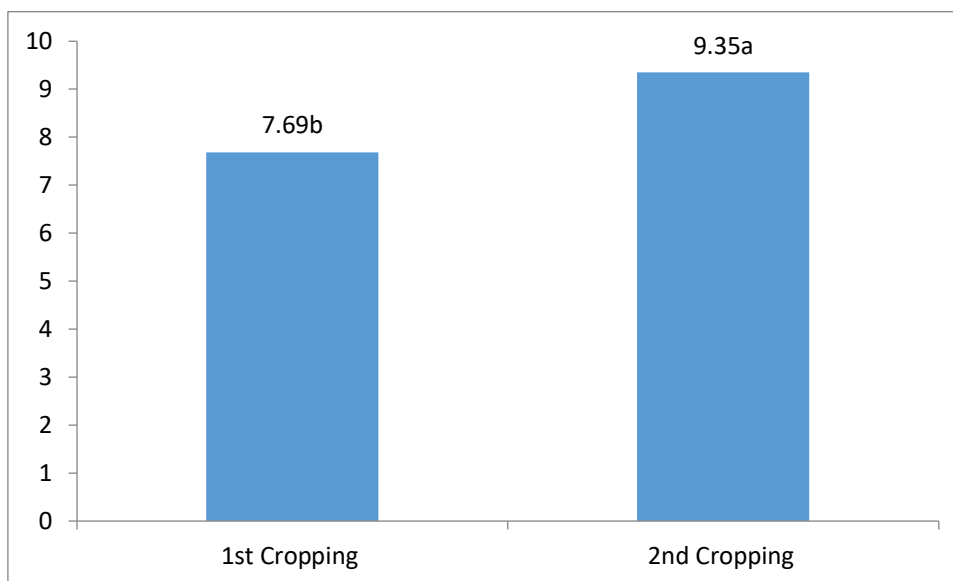


Figure 3. Stem girth (mm) of okra applied with different formulated soil ameliorants as affected by the two cropping seasons.

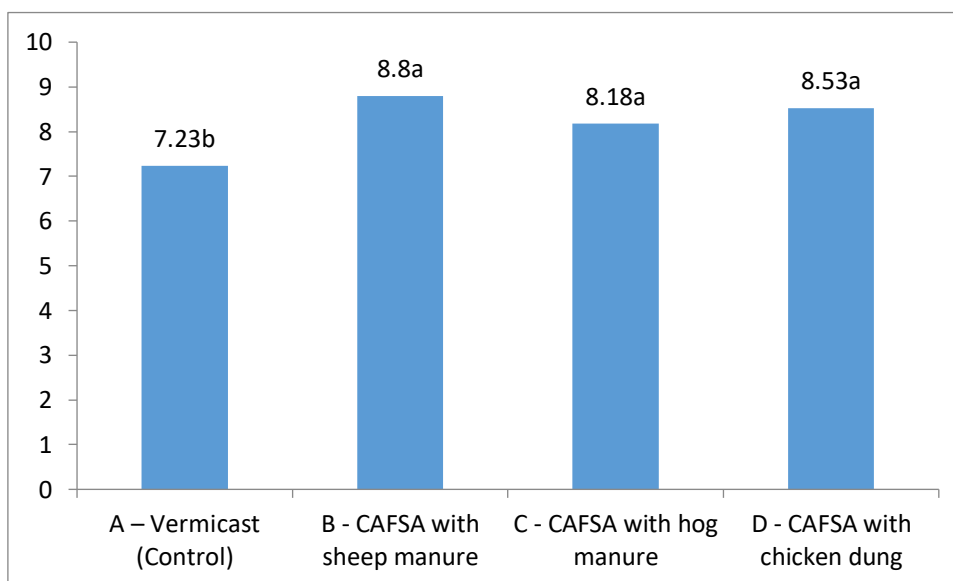


Figure 4. Number of leaves of okra across two cropping seasons as affected by the different formulated soil ameliorants.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 10 of 21

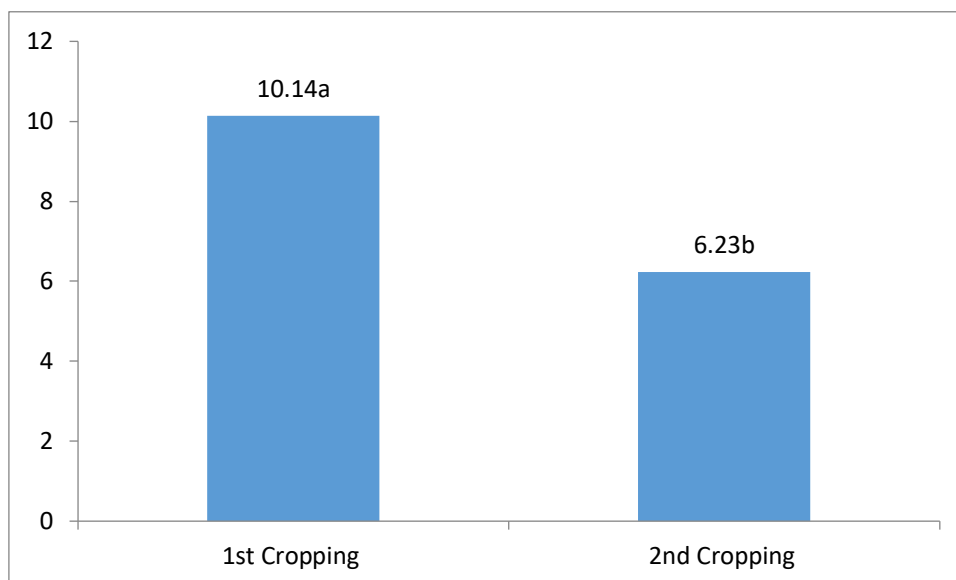


Figure 5. Number of leaves of okra applied with different formulated soil ameliorants as affected by the two cropping seasons.

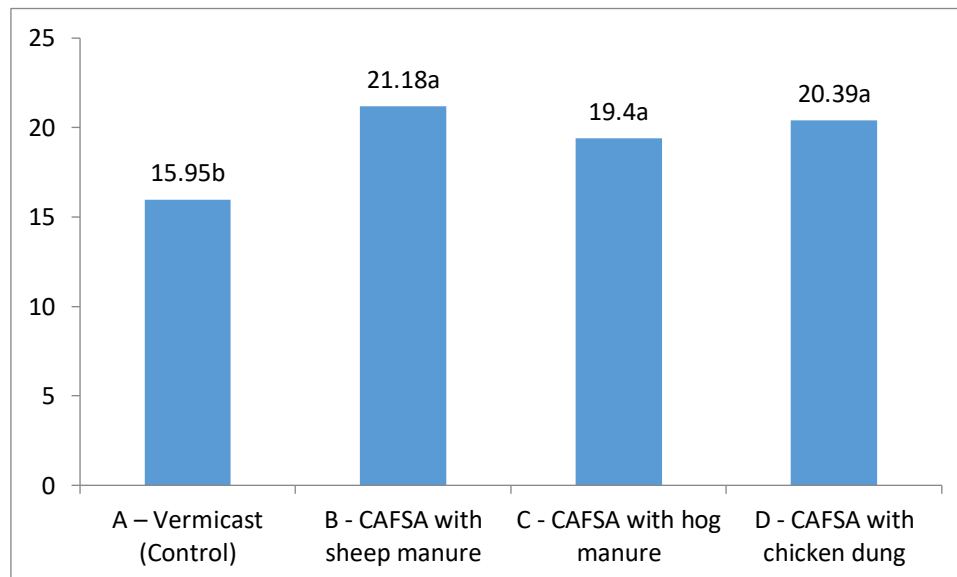


Figure 6. Width of leaf (cm) of okra across two cropping seasons as affected by the different formulated soil ameliorants.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 11 of 21

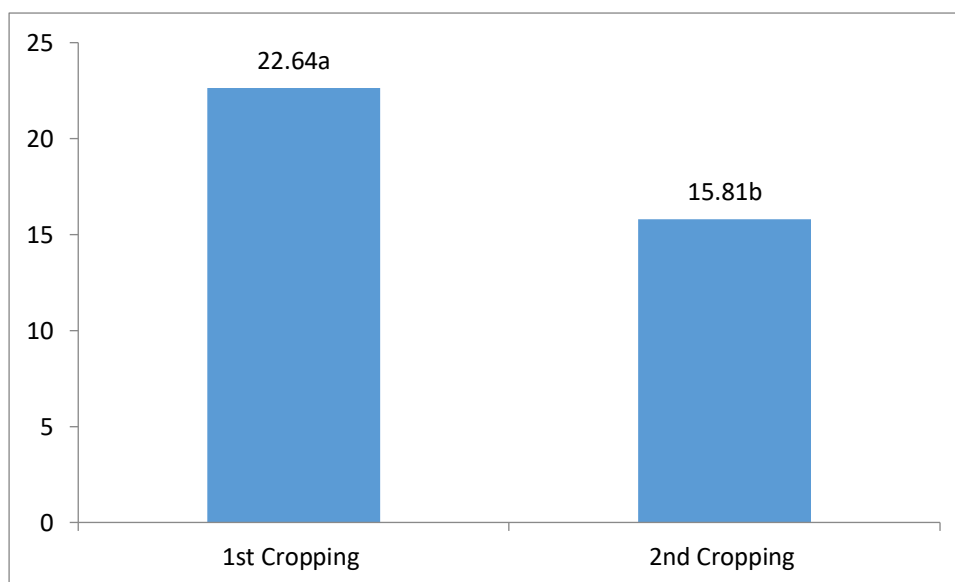


Figure 7. Width of leaf (cm) of okra applied with different formulated soil ameliorants as affected by the two cropping seasons.

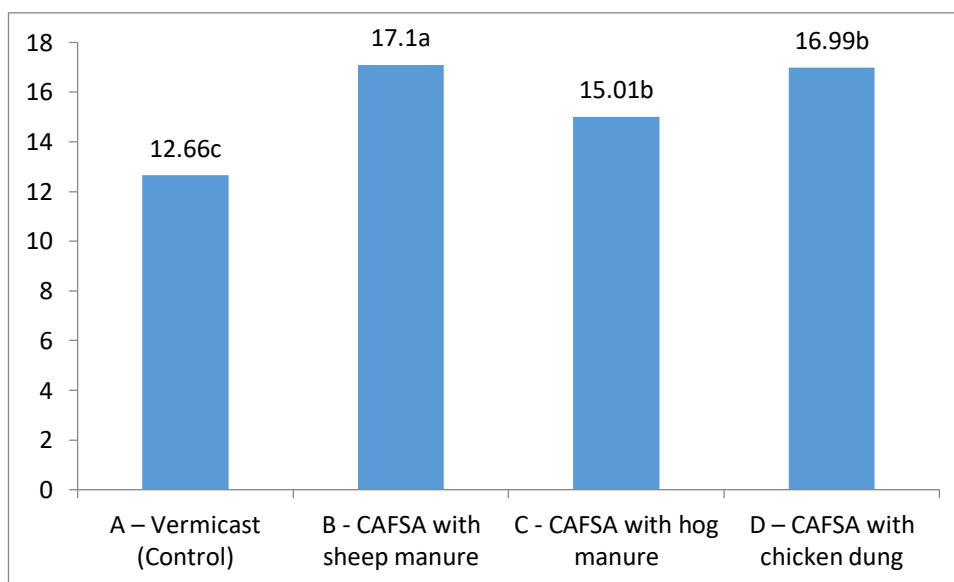


Figure 8. Length of leaf (cm) of okra across two cropping seasons as affected by the different formulated soil ameliorants.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 12 of 21

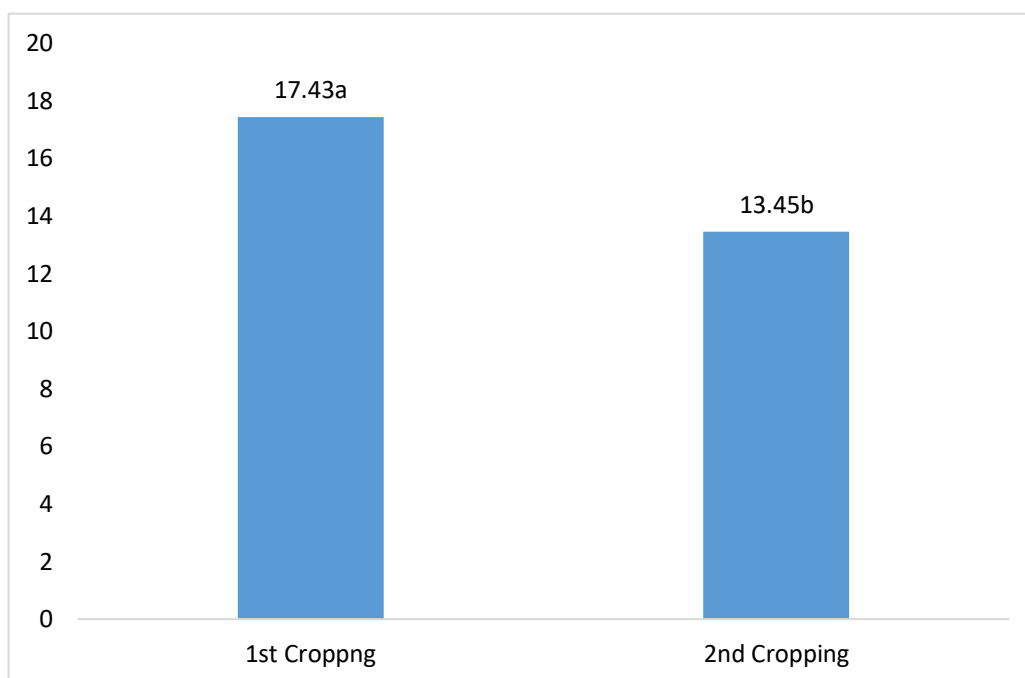


Figure 9. Length of leaf (cm) of okra applied with different formulated soil ameliorants as affected by the two cropping seasons.

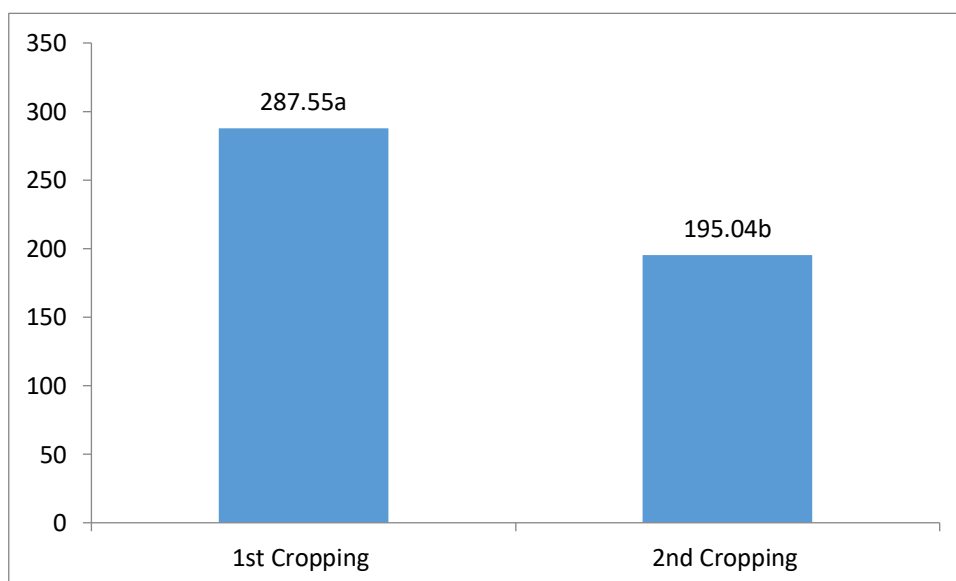


Figure 10. Biomass (g) of okra applied with different formulated soil ameliorants as affected by the two cropping seasons.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 13 of 21

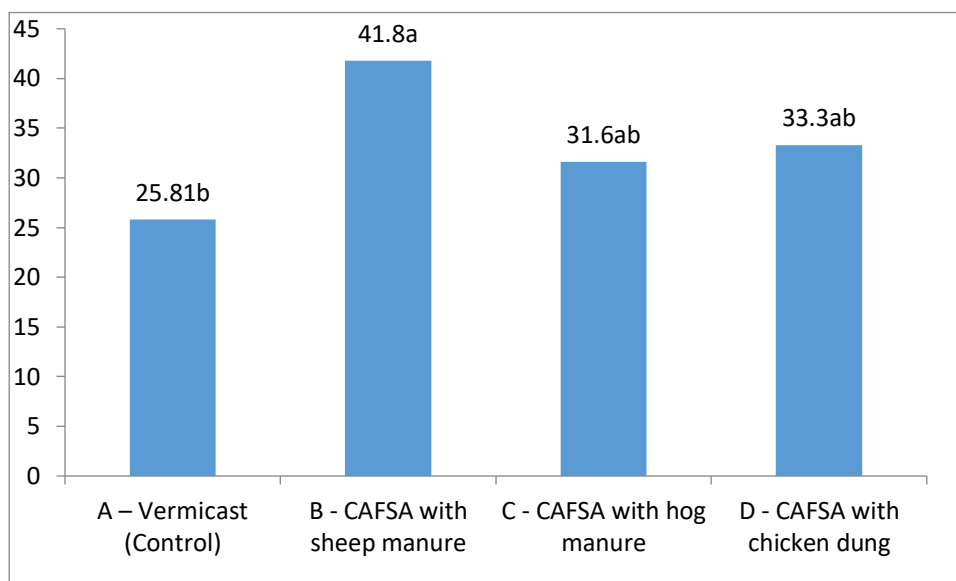


Figure 11. Length of roots (cm) of okra across two cropping seasons as affected by the different formulated soil ameliorants.

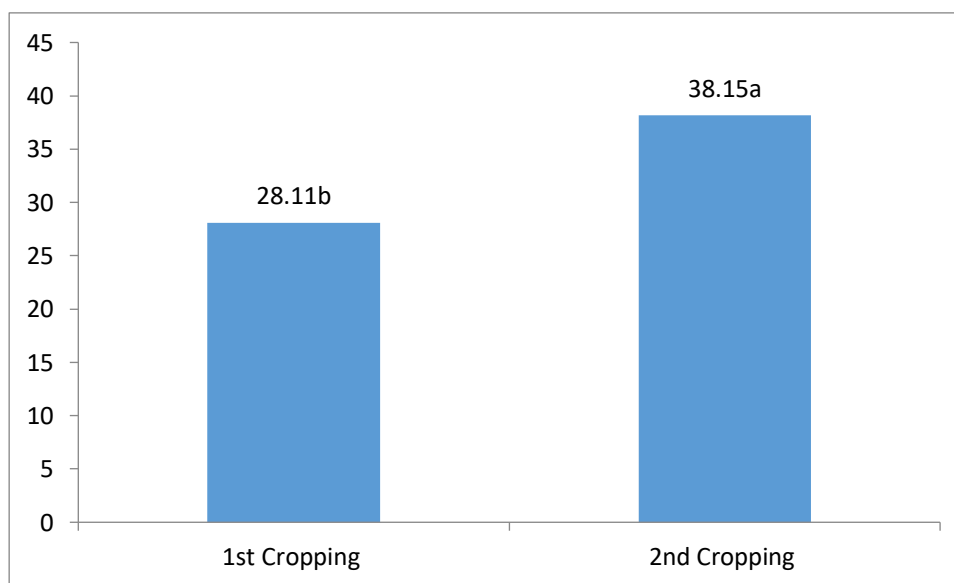


Figure 12. Length of roots (cm) of okra applied with different formulated soil ameliorants as affected by the two cropping seasons.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 14 of 21

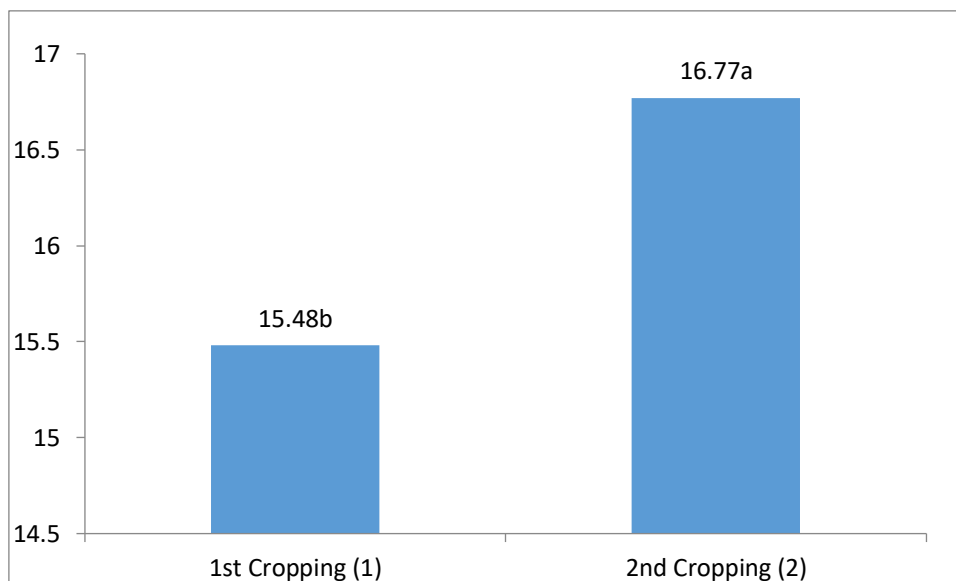


Figure 13. Diameter of fruit (mm) of okra applied with different formulated soil ameliorants as affected by the two cropping seasons.

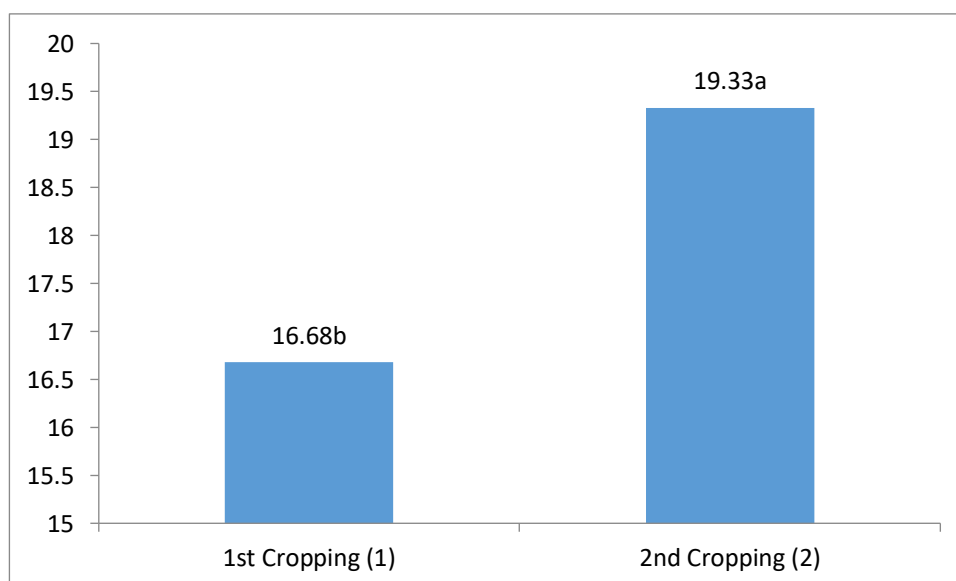


Figure 14. Weight of marketable fruit (g) of okra applied with different formulated soil ameliorants as affected by the two cropping seasons.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 15 of 21

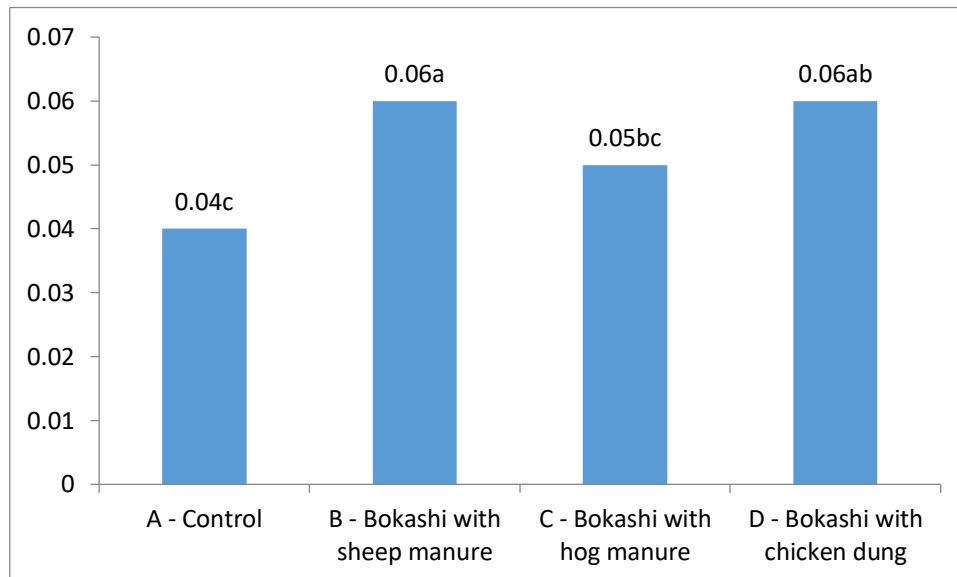


Figure 15. Yield (T/ha) of okra across two cropping seasons as affected by the different formulated soil ameliorants.

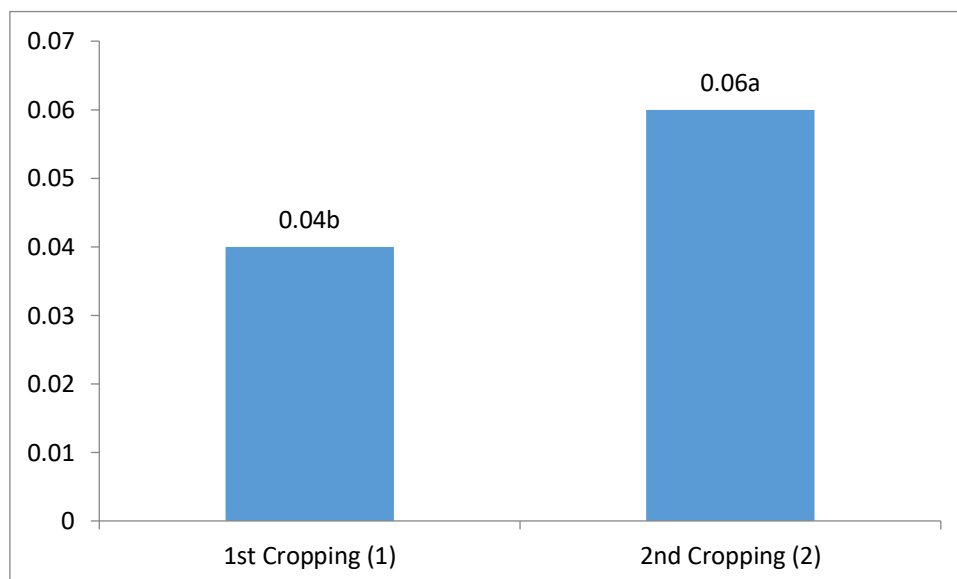


Figure 16. Yield (T/ha) of okra in applied with different formulated soil ameliorants as affected by the two cropping seasons.

	Document Type:	Document Code	RES-F08
	FORM ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	January 26, 2023
	COMPLETED RESEARCH	Page	Page 1 of 21

APPENDICES

Table 1. Plant height (cm) and stem girth (mm) of okra applied with CAPSU Formulated Soil Ameliorants (CAFSU) in different cropping seasons.

Plant Height (cm)					Stem Girth (mm)				
Treatment	Cropping**				Treatment	Cropping**			
CAFSU**	1st Cropping	2nd Cropping	Total	Mean	CAFSU**	1st Cropping	2nd Cropping	Total	Mean
A – Vermicast (Control)	75.80c	56.33e	132.14	66.07	A – Vermicast (Control)	5.87	7.62	13.48	6.74c
B - CAFSA with sheep manure	101.84a	67.65cd	169.50	84.75	B - CAFSA with sheep manure	8.94	10.30	19.24	9.62a
C – CAFSA with hog manure	89.58b	62.55de	152.13	76.06	C - CAFSA with hog manure	7.62	9.42	17.04	8.52b
D - CAFSA with chicken dung	97.29a	59.90de	157.19	78.60	D - CAFSA with chicken dung	8.33	10.05	18.38	9.19ab
Total	364.51	246.44			Total	30.76	37.38		
Mean	91.13	61.61			Mean	7.69b	9.35a		

	Document Type: FORM ISO 9001:2015	Document Code	RES-F08
		Revision No.	00
	Document Title: COMPLETED RESEARCH	Effective Date	January 26, 2023
		Page	Page 2 of 21

Table 2. Number and width (cm) of leaves of okra applied with CAPSU Formulated Soil Ameliorants (CAFSU) different cropping seasons.

Number of leaves					Width of leaves (cm)				
Treatment	Cropping**				Treatment	Cropping**			
CAFSU**	1st Cropping	2nd Cropping	Total	Mean	CAFSU**	1st Cropping	2nd Cropping	Total	Mean
A – Vermicast (Control)	9.09	5.37	14.46	7.23b	A – Vermicast (Control)	18.24	13.66	31.89	15.95b
B - CAFSA with sheep manure	11.07	6.53	17.61	8.80a	B - CAFSA with sheep manure	24.76	17.59	42.35	21.18a
C - CAFSA with hog manure	10.07	6.30	16.37	8.18a	C - CAFSA with hog manure	24.08	14.71	38.79	19.40a
D - CAFSA with chicken dung	10.32	6.73	17.05	8.53a	D - CAFSA with chicken dung	23.48	17.30	40.77	20.39a
Total	40.55	24.93			Total	90.56	63.25		
Mean	10.14a	6.23b			Mean	22.64a	15.81b		

	Document Type: FORM ISO 9001:2015	Document Code	RES-F08
		Revision No.	00
	Document Title: COMPLETED RESEARCH	Effective Date	January 26, 2023
		Page	Page 3 of 21

Table 3. Length of leaves (cm) and biomass (g) of okra applied with CAPSU Formulated Soil Ameliorants (CAFSA) in different cropping seasons.

Length of leaves (cm)					Biomass (g)				
Treatment	Cropping**				Treatment	Cropping**			
CAFSA**	1st Cropping	2nd Cropping	Total	Mean	CAFSA ^{ns}	1st Cropping	2nd Cropping	Total	Mean
A – Vermicast (Control)	14.05	11.26	25.31	12.66c	A – Vermicast (Control)	192.43	167.10	359.53	179.77
B - CAFSA with sheep manure	19.24	14.97	34.20	17.10a	B - CAFSA with sheep manure	380.67	214.71	595.38	297.69
C - CAFSA with hog manure	16.75	13.28	30.03	15.01b	C - CAFSA with hog manure	258.33	209.83	468.17	234.08
D – CAFSA with chicken dung	19.68	14.30	33.98	16.99ab	D – CAFSA with chicken dung	318.77	188.50	507.27	253.63
Total	69.71	53.81			Total	1150.20	780.15		
Mean	17.43a	13.45b			Mean	287.55a	195.04b		

	Document Type:	Document Code	RES-F08
	FORM ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	January 26, 2023
	COMPLETED RESEARCH	Page	Page 4 of 21

Table 4. Length of roots (cm) and weight of roots (g) leaves in okra applied with CAPSU Formulated Soil Ameliorants (CAFSU) in different cropping seasons.

Length of roots (cm)					Weight of roots (g)				
Treatment	Cropping*				Treatment	Cropping ^{ns}			
CAFSA*	1st Cropping	2nd Cropping	Total	Mean	CAFSA ^{ns}	1st Cropping	2nd Cropping	Total	Mean
A – Vermicast (Control)	24.20	27.42	51.62	25.81b	A – Vermicast (Control)	25.33	36.50	61.83	30.92
B - CAFSA with sheep manure	33.27	50.33	83.60	41.80a	B - CAFSA with sheep manure	46.37	32.94	79.31	39.66
C - CAFSA with hog manure	27.03	36.17	63.20	31.60ab	C - CAFSA with hog manure	26.83	30.30	57.13	28.57
D - CAFSA with chicken dung	27.93	38.67	66.60	33.30ab	D - CAFSA with chicken dung	37.17	34.63	71.79	35.90
Total	112.43	152.59			Total	135.70	134.37		
Mean	28.11b	38.15a			Mean	33.93	33.59		

	Document Type:	Document Code	RES-F08
	FORM ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	January 26, 2023
	COMPLETED RESEARCH	Page	Page 5 of 21

Table 5. Diameter of fruit (mm) and length of fruits in okra applied with CAPSU Formulated Soil Ameliorants (CAFSU) in different cropping seasons.

Diameter of fruit (mm)					Length of fruit (cm)				
Treatment	Cropping**				Treatment	Cropping ^{ns}			
Bokashi ^{ns}	1st Cropping (1)	2nd Cropping (2)	Total	Mean	Bokashi ^{ns}	1st Cropping (1)	2nd Cropping (2)	Total	Mean
A - Control	15.48	16.67	32.15	16.07	A - Control	13.07	13.70	26.77	13.39
B - Bokashi with sheep manure	15.20	17.05	32.25	16.12	B - Bokashi with sheep manure	12.95	13.67	26.62	13.31
C - Bokashi with hog manure	15.41	16.61	32.02	16.01	C - Bokashi with hog manure	13.01	13.77	26.77	13.39
D - Bokashi with chicken dung	15.82	16.77	32.59	16.29	D - Bokashi with chicken dung	13.26	13.52	26.78	13.39
Total	61.90	67.10			Total	52.29	54.65		
Mean	15.48b	16.77a			Mean	13.07	13.66		

	Document Type:	Document Code	RES-F08
	FORM ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	January 26, 2023
	COMPLETED RESEARCH	Page	Page 6 of 21

Table 6. Weight of marketable fruit (g) and yield (T/ha) of okra applied with CAPSU Formulated Soil Ameliorants (CAFSU) in different cropping seasons.

Weight of marketable fruit (g)					Yield (T/ha)				
Treatment	Cropping*				Treatment	Cropping**			
Bokashi ^{ns}	1st Cropping (1)	2nd Cropping (2)	Total	Mean	Bokashi**	1st Cropping (1)	2nd Cropping (2)	Total	Mean
A - Control	16.35	16.89	33.25	16.62	A - Control	0.04	0.05	0.08	0.04c
B - Bokashi with sheep manure	16.75	23.33	40.08	20.04	B - Bokashi with sheep manure	0.05	0.07	0.13	0.06a
C - Bokashi with hog manure	16.80	18.09	34.89	17.44	C - Bokashi with hog manure	0.04	0.05	0.09	0.05bc
D - Bokashi with chicken dung	16.81	19.02	35.83	17.91	D - Bokashi with chicken dung	0.05	0.06	0.11	0.06ab
Total	66.71	77.33			Total	0.18	0.24		
Mean	16.68b	19.33a			Mean	0.04b	0.06a		